

Functional Programming - 2|Lecture

- Principles of Functional Programming
- Maps
- Filter
- Zip
- Reduce
- Args and Kwargs

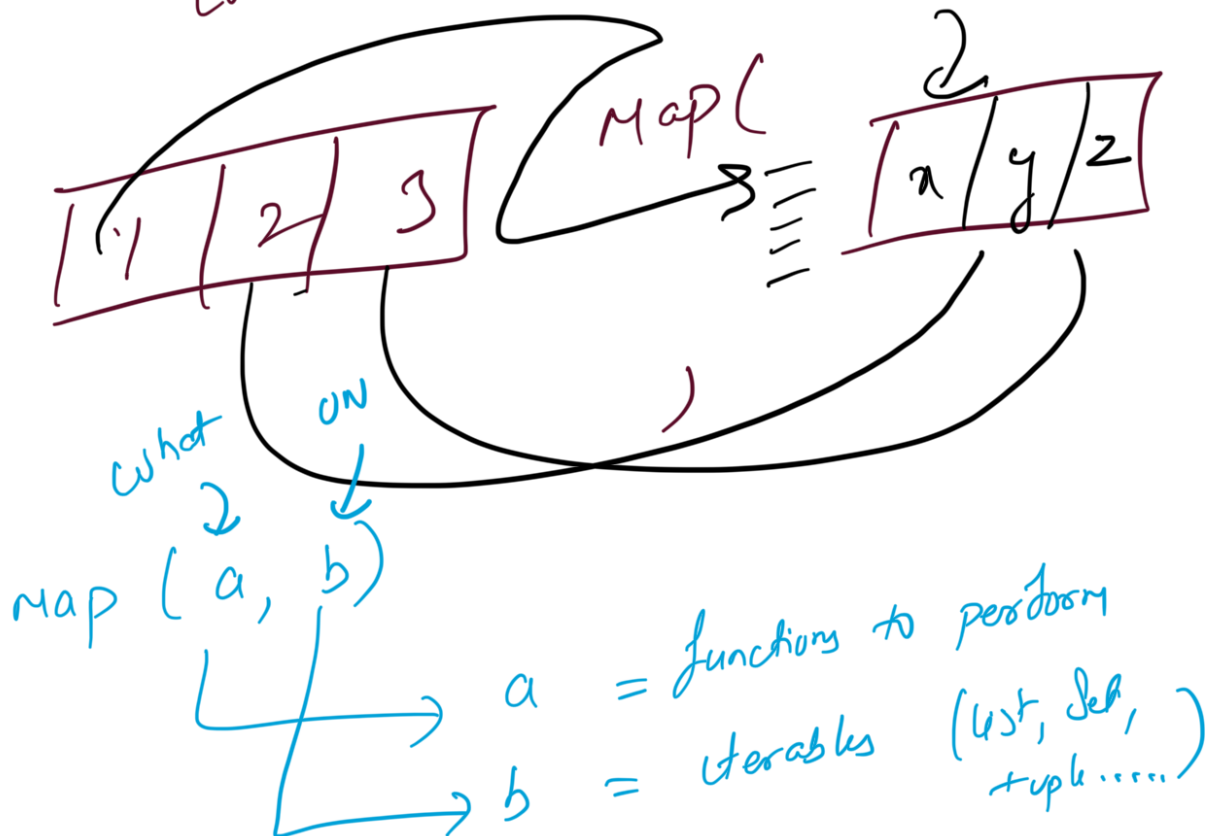


Principles of FP:-

- ① Data should be separate from mutations
- ② Treat variables as immutable (unchangeable)
- ③ Treat functions as FCC
 - ↳ HOF
 - ↳ Any f(a)

① Map()

↳ It applies a function to every item in a iterable (list, tuple etc) & returns a map object (which can be converted to a list/tuple)



$$a = [1, 2, 3, 4, 5]$$

$$\Downarrow$$

$$[1, 4, 9, 16, 25]$$

② Filter
↳ filter out elements from a list on basis of some conditions.

$$a = [1, 2, 3, 4, 5]$$

$$f(x) = x/2 = 0$$

$$[2, 4]$$

(Note: In the original image, the elements 2 and 4 in the list and the result list are circled, and the condition $x/2 = 0$ is written below the list.)

③ Zip : It combines 2 or more iterables element wise into pairs

$$\text{Zip}(\text{iter 1}, \text{iter 2}, \dots)$$

$$\begin{aligned} \text{name} &= ["A", "B", "C"] \\ \text{marks} &= [85, 90, 80] \end{aligned}$$

(Note: In the original image, the first elements 'A' and 85 are circled, and arrows point from them to each other.)

$$\text{zip}(\text{name}, \text{marks})$$

****kwargs**

↳ allows a function to accept a variable
no of keyword arguments

`sum(x=1, y=2, args=" ")`

Order of passing arguments

`P → Arg → K → Kwargs`